WORKED SOLUTION

STEP II 2020, Question 4

Pure • Triangle Inequality

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Everything rests on the triangle inequality: each side is strictly shorter than the sum of the other two. In fact only one inequality ever needs checking, the one for the *longest* side, because the other two hold automatically. For the converse, build the triangle with compasses: put down one side and swing two circles from its ends.

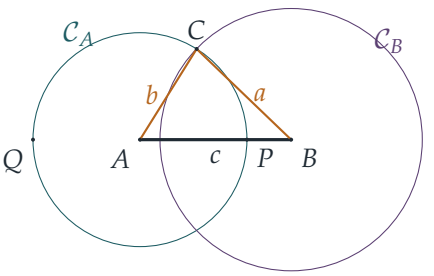
Solution

Part (i)

Label the vertices A, B, C with $BC = a$, $CA = b$, $AB = c$. The straight segment AB is the shortest route from A to B, so the route $A \to C \to B$, of length $b + a$, is at least c. It can't equal c: that happens only when C lies on the segment AB, and then there is no triangle. Hence $c < a + b$. The same argument for the other two pairs of vertices gives $a < b + c$ and $b < a + c$.

Part (ii)

Draw a segment AB of length c. Draw the circle $\mathcal{C}_A$ with centre A and radius b, and the circle $\mathcal{C}_B$ with centre B and radius a. If they meet at a point C off the line AB, then ABC is a triangle with $CA = b$, $CB = a$ and $AB = c$.



Why must they meet? Let P and Q be the points where $\mathcal{C}_A$ crosses the line AB, with P on the same side of A as B. Then

$$PB = |c - b|, \qquad QB = c + b.$$

Now $|c - b| < a$, because $c - b < a$ and $b - c < a$ are two of the given inequalities; and $c + b > a$ is the third. So as a point travels round the upper half of $\mathcal{C}_A$ from P to Q, its distance from B changes continuously from a value below a to a value above a, and therefore equals a somewhere. That point lies on both circles, and it is not on the line AB, since the only points of $\mathcal{C}_A$ on that line are P and Q, whose distances from B are not a. This is the required point C, so a triangle with sides a, b, c exists.

Part (iii)

Parts (i) and (ii) together say that three positive numbers are the sides of a triangle exactly when each is less than the sum of the other two. Write ℓ for the largest of the three. If ℓ is less than the sum of the other two, the other two inequalities follow at once, since each of the smaller numbers is at most ℓ and the remaining number is positive. So we only ever need to test the largest length.

(A) Always. All three lengths are positive. Using $c < a + b$,

$$c + 1 < a + b + 1 < (a + 1) + (b + 1),$$

and the other two inequalities follow in the same way.

(B) Sometimes but not always. For an equilateral triangle the three lengths are $1, 1, 1$, which do form a triangle. But take $a = b = 1$ and $c = \frac{1}{10}$, which is a genuine triangle since $1 < 1 + \frac{1}{10}$. The lengths become 1, 10, $\frac{1}{10}$, and $10 > 1 + \frac{1}{10}$, so no triangle is possible.

(C) Never. The three lengths are symmetric in a, b, c, so we may label so that $a \geq b \geq c$. Then

$$|c - a| = a - c = (a - b) + (b - c) = |a - b| + |b - c|.$$

The largest length equals the sum of the other two, so the strict inequality fails. (If two of a, b, c are equal, one of the lengths is 0, which fails too.)

(D) Always. The set $\{a^2 + bc,\ b^2 + ca,\ c^2 + ab\}$ doesn't change when we swap any two of a, b, c. (Swapping a and b, for instance, gives $b^2 + ac$, $a^2 + cb$, $c^2 + ba$.) So it is enough to prove one of the three inequalities, say $a^2 + bc < (b^2 + ca) + (c^2 + ab)$, for every triangle. The difference is

$$\begin{aligned}(b^2 + ca) + (c^2 + ab) - (a^2 + bc) &= (b^2 - bc + c^2) + a(b + c) - a^2 \\ &= (b - \tfrac{1}{2}c)^2 + \tfrac{3}{4}c^2 + a(b + c - a).\end{aligned}$$

The first bracket is positive because $c > 0$, and $a(b + c - a) > 0$ because $a > 0$ and $a < b + c$. So the difference is positive. The other two inequalities follow by relabelling, and all three lengths are clearly positive, so they always form a triangle.

Part (iv)

The roles of a, b, c are interchangeable, so we may label so that $a \geq b \geq c$. All three of $f(a)$, $f(b)$, $f(c)$ are positive. Because f is increasing, $f(a) \geq f(b) \geq f(c)$, so by the remark at the start of part (iii) we only need

$$f(a) < f(b) + f(c).$$

Ask yourself How can the condition on $f(x)/x$ help compare $f(a)$ with $f(b) + f(c)$?

It lets us bound $f(b)$ and $f(c)$ *below* by multiples of $f(a)$. Since $f(x)/x$ decreases as x increases, and $a \geq b$,

$$\frac{f(b)}{b} \geq \frac{f(a)}{a}, \qquad \text{so} \qquad f(b) \geq \frac{b}{a} f(a),$$

with equality allowed only when $b = a$. In the same way $f(c) \geq \frac{c}{a} f(a)$. Adding,

$$f(b) + f(c) \;\geq\; \frac{b + c}{a} f(a) \;>\; f(a),$$

where the last step uses $b + c > a$ (from the triangle) and $f(a) > 0$. So the largest of the three new lengths is less than the sum of the other two, and $f(a)$, $f(b)$, $f(c)$ are the sides of a triangle.

Checks

- For (D), try $a = 2$, $b = 3$, $c = 4$: the lengths are 16, 17, 22, and $22 < 16 + 17$. The identity, relabelled for the longest length $c^2 + ab$, predicts the gap $(a^2 - ab + b^2) + c(a + b - c) = (4 - 6 + 9) + 4 \cdot 1 = 11$, and indeed $16 + 17 - 22 = 11$.
- For (C), the triangle $3, 4, 5$ gives $1, 1, 2$: flat, exactly as predicted.
- A computer check on 200 000 random triangles confirmed (A), (D) and (C) (always, always, never), found (B) holding for about 40% of them, and confirmed part (iv) for $f(x) = \sqrt{x}$, $\ln(1 + x)$, $x/(1 + x)$, $x^{0.3}$ and $1 - \mathrm{e}^{-x}$.

Common slips

- Answering (B) with "never" or "always" after testing one triangle: a "sometimes" answer needs one example of each kind.
- In (D), checking only one inequality without explaining why the other two follow: the symmetry (or the longest-side remark) is what justifies it.

- In (iv), writing $f(a)/a < f(b)/b$ when a might equal b; the inequality is then $\leq$, and the strictness has to come from $b + c > a$.

Beyond STEP

The triangle inequality is one of the three axioms of a *metric*, the abstract notion of distance at the heart of first-year analysis and topology. Part (iv) is a classic fact there: if d is a metric and f is increasing with $f(0) = 0$ and $f(x)/x$ decreasing, then $f \circ d$ is a metric too; this is why $d/(1 + d)$ is a standard way to make a bounded metric. Bartle and Sherbert, *Introduction to Real Analysis*, is a good place to meet these ideas.

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